Experiment 5.

OBJECT

To study the variation of semiconductor resistance with temperature and hence determine the band gap of semiconductor in the form of reverse biased P.N. junction diode.

APPARATUS

A semi conductor diode, a micrommeter a battery, a thermometer, oven or calorimeter with stirrer and carbon tetra chloride (CCl_4) liquid.

FORMULA

A graph is plotted between log I_s and $\left(10^3/T\right)$ that comes out to be a straight line. Its slope is determined.

Band gap, ΔE, in electron volts, is given by

$$\Delta E = \frac{slope\ of\ the\ line}{5.036}\ electron\ volts \qquad ...(1)$$

THEORY

In a semiconducter, there is an energy gap between its conduction and valence band. For conduction of electricity, a certain amount of energy is to be given to the electron so that it goes from the valence band to the conduction band. The energy so needed is the measure of the energy gap, ΔE, between two bands.

When a P-N junction is reverse biased then current is due to minority carriers whose concentration is dependent on the energy gap, ΔE. The reverse current, I_s (saturated value) is a function of the temperature of the junction diode for small range of temperature, the relation is expressed as.

$$\log I_s = log\left[A\ e\ N_n\ N_p\left(\frac{V_n}{p_p}+\frac{V_p}{n_n}\right)\right] - e\frac{\Delta E}{kT} \qquad .....(2)$$

Where

n_n = density of electrons in n type material

p_p = density of holes in P type material

V_n = Velocity of electrons

V_p = Velocity of holes.

A = Area of the junction

k = Boltzmann constant

T = Absolute temperature of junction

$$N_n = \frac{2(2\pi m_n kT/e)^{3/2}}{h^3} \quad .....(3)$$

$$N_p = \frac{2(2\pi m_p kT/e)^{3/2}}{h^3} \quad .....(4)$$

m_n is mass of an electron and m_p is the effective mass of the hole. So for small range of temperature, relation (2) can be written as

$$\log I_s = constant - 5.036\Delta E\left(\frac{10^3}{T}\right) \quad .....(5)$$

Where temperature T is in kelvin, ΔE is in electron volts. A graph between log I_S and $\left(10^3/T\right)$ is plotted which comes out to be a straight line. The slope of this line will be 5.036 ΔE, giving the value of band gap for the semiconductor i.e.

$$\Delta E = \frac{Slope}{5.036} \quad .....(6)$$

PROCEDURE

1. Connect the diode (Germanium or Silicon) in series with a microammeter and a battery as shown in Fig 1 given below.
2. Heat carbon tetrachloride liquid in a colorimeter up to a temperature of 70°C if germanium diode is taken or about 120°C if silicon diode is taken.
3. Put this calorimeter in a wooden box in which cotton is placed. This lowers the rate of cooling. Now put the diode in the heated liquid. Put the bulb of a thermometer in contact with the diode.

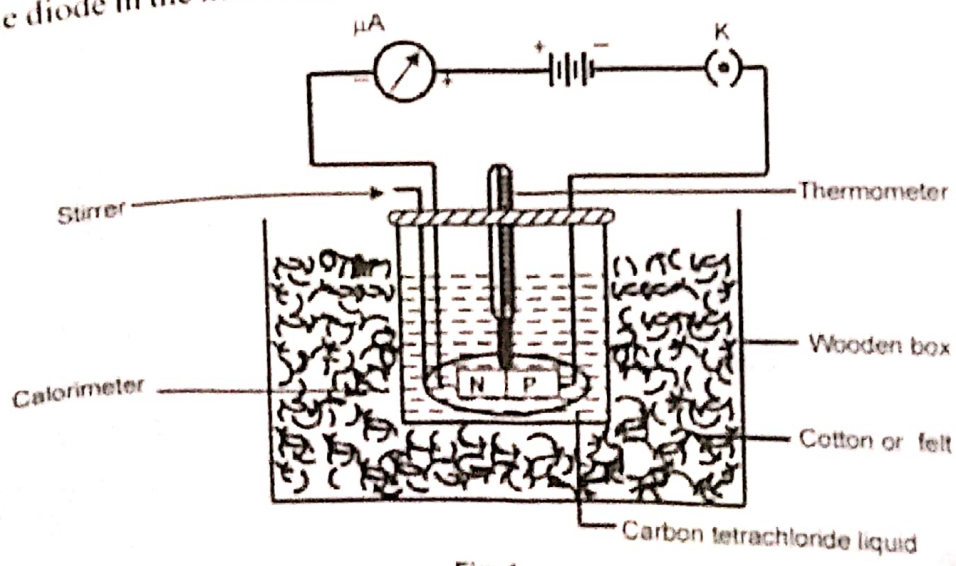

Fig. 1

4. Stir the liquid (by a stirrer) continuously so that temperature remains uniform.
5. The key is closed and the temperature is noted for various values of current as the diode cools down.
6. Now plot a graph betwen log I_s and $10^3/T$. It is a straight line as shown in figure.
7. Detemine the slope of this line dy/dx and from this slope detemine the band gap ΔE with the help of relation (6)

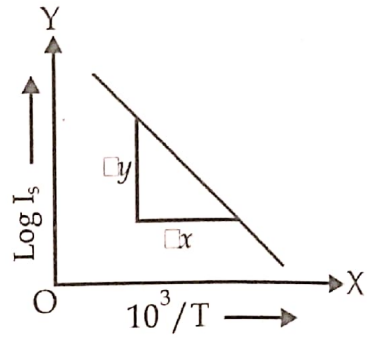

Fig.2

OBSERVATIONS :

S.No.	Temperature T (in °C)	Temperature T (in K)	Current I_S (in μA)	$10^3/T$	Log $_{10}$ I_s
1.					
2.					
3.					
4.					
5.					
6.					
7.					

CALCULATIONS

From the graph, slope of the line $\frac{dy}{dx}$ = ...

∴ energy gap $\Delta E = \frac{Slope}{5.036}$ =.....eV

RESULTS

The energy band gap for given semiconductor = ... eV

Standard value for Germaiun = 0.72 eV

Standard value for Silicon = 1.10 eV

Percentage error = ...%

PRECAUTIONS

1. The diode should be reversed biased.
2. Do not heat the liquid beyond 70°C in case of germanium diode and 120°C in case of silicon diode.
3. The cooling rate should be slow so that the diode gets necessary time to attain the temperature of the liquid.
4. Liquid should be continuously stirred.
5. Radiation losses should be minimised as far as possible.

VIVA–VOCE

1. What is a band gap?

Ans. This is energy gap between the conduction and valence bands of a semiconductor (or insulators)

2. What is band gap in a good condcutor?

Ans. There is no band gap as the two bands overlap in the good conductors. There is very little band gap in some conductors.

3. How is reverse current produced across a P-N junction and on what factors does it depend?

Ans. When a P-N junction is reversed biased. then current is due to minority carriers whose concentration is dependent on energy gap or band gap.

4. What are semiconductors?

Ans. Semiconductors are those materials whose resistivity lies between those of conductors and insulators i.e. it is of the order of 10^4 to 10^{-1} ohm-meter.

5. What is intrinsic and extrinsic semiconductors?

Ans. A pure semiconductor like germanium or silicon is called a pure semiconductor. When some pentavalent on trivalent impurity is added to a semiconductor in order to increase its conductivity, it is called the extrinsic semiconductor.

6. What are N-type and P-type semiconductors?

Ans. Pure semiconductors are 4th group elements. If a trivalent impurity like Al, B, Ga, In etc. is added to it, the semiconductor is called P-type. When a pentavalent impurity, like P, As, Sb, Bi etc. is added to it, it is called the N-type semiconductor.

7. What do you mean by forward and reverse biasing of junction diode?

Ans. If the positive terminal of the battery is connected to the P-side and negative terminal to the N-side of the junction that is called the forward biasing. On reversing the connections either of battery or of the diode, it is said to be reverse biased.